

Lecture 24: Maximum Likelihood Estimation

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- 1 Maximum likelihood::: {.callout-note icon=false}## Definition: Likelihood / MLEFor data $x_1, \dots, x_n$ with parameter θ : $L(\theta) = \prod_i f(x_i; \theta)$, and

$$\hat{\theta}_{\text{MLE}} = \arg \max_{\theta} L(\theta) = \arg \max_{\theta} \log L(\theta).$$

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- 1.1 Reading & HomeworkReading: §9 (Bertsekas & Tsitsiklis, *Introduction to Probability*).Due today: Homework 11 (Gradescope). 1

—title: “Lecture 24: Maximum Likelihood Estimation”author: “Jamie Haddock”format: revealjs: output-file: Lecture24_slides slide-number: true preview-links: auto logo: figs/hmc.png css: input/slides.css incremental: true smaller: true code-fold: true embed-resources: true html: code-fold: true embed-resources: true pdf: documentclass: article toc: true number-sections: true geometry: - top=1in - left=1in - bottom=1in - right=1informat-links: falsejupyter: python3filters: - input/remove-pause.luaexecute: echo: true eval: true—

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```
from scipy.optimize import minimize_scalarfrom scipy.stats import poissonlam_true, n = 4.5, 40data = rng.poisson(lam_true, size=n)ns = [5, 20, 100, 500, 2000]means_by_n = [np.mean([rng.poisson(lam_true, size=n_).mean() for _ in range(n_)])
```

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